

SAM10 AWS INFRASTRUCTURE DESIGN — SYSTEM PROPOSAL

Phase 1: Project System Idea & Infrastructure Requirements Mapping

Student Name: Robert Angel B. Santiago	Student ID: 23-1360-266	Course Section: BSIT-4-A	Instructor: Romack L. Natividad, MIT
--	-------------------------	--------------------------	--------------------------------------

1. CHOSEN SYSTEM IDEA & PURPOSE

MediCare Direct is an integrated outpatient clinic management and patient appointment scheduling system. The platform provides an online portal for patients to search doctor availability, schedule appointments, and view lab test notifications. Internally, medical staff and physicians utilize the system to manage daily clinical rosters and retrieve electronic health records (EHR). The major infrastructure requirements include a publicly accessible web interface, a isolated application backend for processing appointment workflows, an encrypted database store for sensitive patient medical data, strict subnet isolation, port-level firewall security, and multi-factor authenticated IAM administrative controls.

2. SYSTEM REQUIREMENTS TO AWS SOLUTION MAPPING

System Requirement	Proposed AWS Service	Technical Specification & Purpose
Public Web Portal	Amazon EC2 (Web Tier)	t3.micro running Amazon Linux 2023 & NGINX reverse proxy for public SSL/TLS termination.
Application Processing	Amazon EC2 (App Tier)	t3.small running Node.js Express REST API for scheduling business logic & EHR processing.
Persistent Data Store	Amazon EBS (Data Tier)	100 GB gp3 SSD volume encrypted with AWS KMS (AES-256) storing PostgreSQL clinical data.
Network Isolation	Amazon VPC & 3 Subnets	10.0.0.0/16 VPC divided into Public (10.0.1.0/24), Private App (10.0.2.0/24), Data (10.0.3.0/24).
Network Security	AWS Security Groups	Stateful firewalls using Security Group ID chaining (sg-WebTier -> sg-AppTier -> sg-DBTier).
Administrative Access	AWS IAM & SSM	Role-Based Access Control (RBAC) with MFA enforcement & SSM Session Manager (no open SSH).

3. NETWORK TOPOLOGY & ARCHITECTURAL JUSTIFICATION

Target Region & CIDR: Deployed in AWS Singapore (ap-southeast-1) within a 10.0.0.0/16 VPC. Public web servers reside in Subnet 10.0.1.0/24 with an Internet Gateway route (0.0.0.0/0 -> IGW). Application processing (10.0.2.0/24) and PostgreSQL database storage (10.0.3.0/24) are strictly isolated in private subnets with no direct Internet routes.

Architectural Alignment: (1) Segmentation isolates data from public access; (2) Least Privilege restricts ports to exact source security groups; (3) Controlled Connectivity enforces unidirectional traffic flow; (4) Scalability CIDRs leave 250+ available IPs per subnet for Auto Scaling; and (5) Simplicity eliminates unnecessary managed service overhead.